INCIDENT RESPONSE

& STANDARD SEVERITY LEVELS

A Practical, Decision-Focused Guide for Alert Handling and Incident Management

AGENDA

01 **Objective**
Purpose and goals of this framework

02 **Core Principles**
Distributed ownership, bias toward action

03 **PREPARE Phase**
Alerting, on-call readiness, runbooks, tooling

04 **RESPOND Phase**
Detect → Triage → Mitigate → Resolve

05 **LEARN Phase**
Post-incident review and remediation

06 **Severity Levels**
P1 Critical through P4 Low definitions

07 **Quick Decision Checklist**
Responder reference guide

08 **Key Takeaways**
Summary and final reminders

OBJECTIVE

Why This Framework Exists

This framework provides engineers, on-call responders, and incident commanders with a clear, actionable guide for detecting, responding to, and learning from technical incidents. It standardizes how incidents are declared, triaged, and resolved. Reducing confusion, cutting down recovery time, and building organizational resilience.

Ensure Fast Detection

Minimize time-to-detect through actionable, accurate alerts

Enable Coordinated Response

Remove ambiguity with clear roles, channels, and escalations

Reduce Recovery Time

Bias toward speed and action, not perfect diagnosis

Drive Continuous Improvement

Learn from every incident to prevent recurrence

CORE PRINCIPLES

PRINCIPLE 1

Distributed Ownership

Any engineer can declare an incident or raise its severity. Do not wait for permission. Ownership belongs to everyone.

PRINCIPLE 2

Bias Toward Overreaction

If unsure between two severity levels, choose the higher one. It is easier to de-escalate than to recover lost time.

PRINCIPLE 3

Action Over Precision

Restoring service is the top priority. Precise impact measurement comes later. Speed of mitigation always wins.

PHASE 1: PREPARE

Before Anything Breaks

Alerting Standards

- Actionable : clear next step
- Accurate : low false positives
- Timely : detect real user impact
- Must include runbook + owner

On-Call Readiness

- Primary + secondary rotation
- Clear escalation path
- Acknowledge within SLA (5 min)
- Own incident until explicit handoff

Runbooks

- Step-by-step failure instructions
- How to verify the issue
- How to mitigate quickly
- When to escalate further

Incident Tooling

- Incident tracking system
- Comms channel (Slack/Teams)
- Status updates (internal + ext.)
- Standardized across teams

PHASE 2: RESPOND

When an Incident Happens — Overview Flow

DETECT → **TRIAGE** → **MITIGATE** → **RESOLVE**

DETECT	TRIAGE	MITIGATE	RESOLVE
Create visibility, initiate coordinated response	Assess impact, determine urgency	Stop or reduce user impact, speed > elegance	Restore normal, stable operation

Key Decision Rules for Responders

- ✓ If debating between severity levels → choose the HIGHER one
- ✓ Escalate if you cannot identify cause within 10–15 min (P1/P2)
- ✓ Prioritize stopping impact, do not wait for root cause
- ✓ Communicate every 15–30 min during active P1/P2 incidents

RESPOND: DETECT & TRIAGE

DETECT

1. Alert fires / User report / Engineer observes anomaly
2. Create Incident Record immediately if none exists
3. Required: Title, Time detected, Affected service(s), Reporter
4. Assign initial severity
5. Create dedicated comm channel (Slack/Teams)
6. Assign Incident Commander if severity ≥ P2
7. Notify on-call, relevant teams, leadership for P1

TRIAGE

Key Questions to Answer:

? Is this customer-facing?

? How many users are impacted?

? Is the issue spreading or contained?

? Is there a workaround?

Escalation Triggers:

⚡ Cannot ID cause within 10–15 min (P1/P2)

⚡ Impact is increasing

⚡ Critical systems involved

RESPOND: MITIGATE & RESOLVE

MITIGATE · Speed > Elegance

Rollback
Revert recent deploy or config change

Traffic Control
Rate limit, feature flags, redirect

Failover
Switch to backup systems or regions

Graceful Degradation
Disable non-critical features, serve cached

Restart / Reset
Services, jobs, or infra components

⚠ Do not wait for root cause to act · Prefer reversible actions

RESOLVE · Restore Normal Operation

1. **Verify Recovery**
   Metrics normal, errors resolved, impact gone
2. **Remove Temporary Fixes**
   Re-enable features carefully, monitor regression
3. **Confirm Stability Window**
   Observe system 15–60 minutes post-fix
4. **Close Incident Record**
   Log timeline, actions, current status
5. **Communicate Resolution**
   Notify stakeholders, summarize impact, schedule PIR

PHASE 3: LEARN

After the Incident — Prevent Recurrence & Improve Response

Post-Incident Review (PIR)

Required for all P1

Deliverables:

- Root cause analysis
- Timeline of events
- What worked / what didn't
- Action items with owners + deadlines

Focus Areas:

→ Detection gaps
→ Response delays
→ Alert quality
→ Coordination issues

Remediation & Follow-Through

Short-Term Fixes

✓ Patch identified bugs
✓ Improve alert coverage
✓ Update runbooks

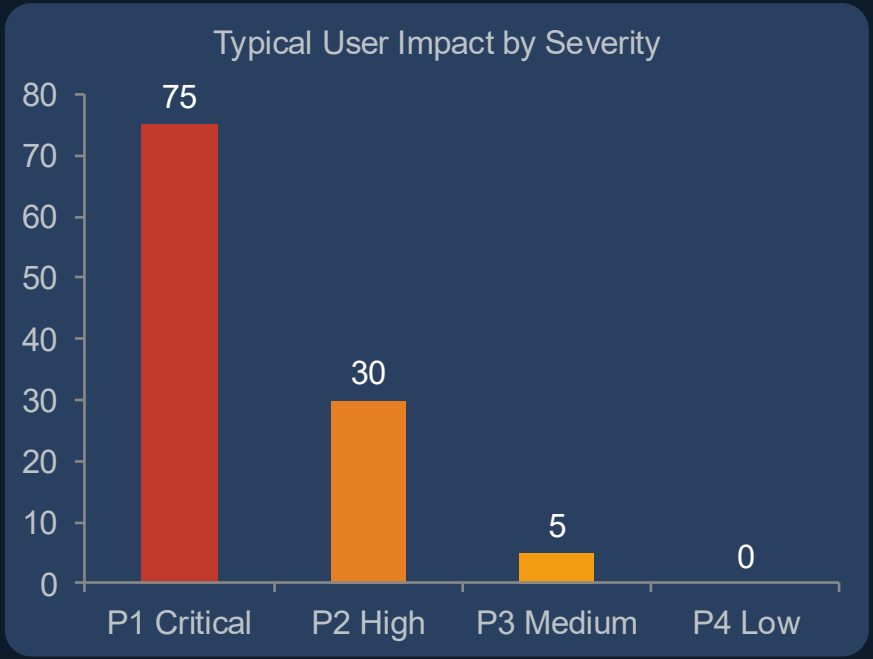
Long-Term Improvements

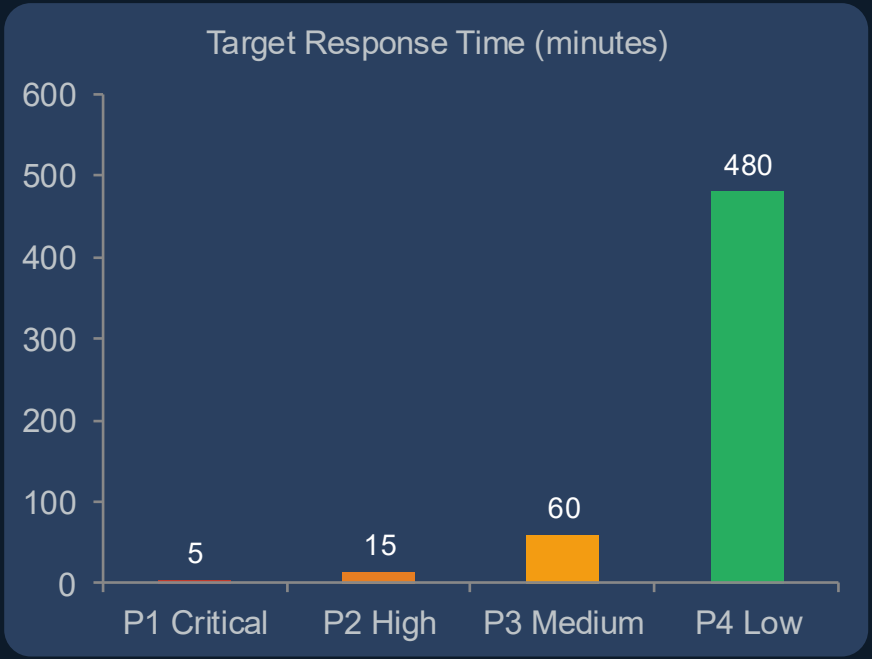
✓ Architectural changes
✓ Reliability investments
✓ Track action items to completion
✓ Review trends across incidents

STANDARD SEVERITY LEVELS

P1 CRITICAL	**Complete or near-complete outage of critical functionality** ≥50% users impacted · Core business functionality unavailable · Data loss or corruption	**IMMEDIATE : Full team mobilized** E.g.: Entire site is down | Payments failing globally | Auth system unavailable
P2 HIGH	**Major degradation with significant user impact** 10–50% users impacted · Key feature partially unavailable · No easy workaround	**URGENT : IC assigned, leadership notified** E.g.: Checkout failing intermittently | API latency extremely high | Mobile app crashing widely
P3 MEDIUM	**Moderate issue with limited impact** <10% users impacted · Non-critical feature broken · Workaround exists	**SCHEDULED : Team responds within hours** E.g.: Search results delayed | Notifications not sending | Minor UI feature broken
P4 LOW	**Minimal impact or internal issue** No significant user impact · Cosmetic or edge-case issues	**PLANNED : Addressed in normal workflow** E.g.: Minor UI bug | Logging issue | Internal dashboard incorrect

SEVERITY IMPACT & RESPONSE TIME





QUICK DECISION CHECKLIST

When Alert Fires

- Is there an incident already?
- If not, create one immediately
- Assign severity (err high)
- Notify on-call + stakeholders

Within First 10 Minutes

- Assess full impact scope
- Escalate if unclear or growing
- Begin mitigation actions

During Incident

- Prioritize stopping user impact
- Document all actions in real time
- Communicate every 15–30 min (P1/P2)

Before Closing

- Confirm full recovery
- Monitor for stability
- Send resolution update to stakeholders

KEY TAKEAWAYS

Do not wait for certainty to act

Speed of mitigation matters more than perfect diagnosis. Bias toward action.

Incidents are a team sport

Any engineer can declare an incident. Distributed ownership, not gatekeeping.

Over-communicate, not under

Update stakeholders every 15–30 min during P1/P2. Silence creates panic.

Always err on higher severity

Easier to de-escalate a P1 than to recover from a misclassified P3.

Prepare before incidents happen

Actionable alerts, on-call rotation, runbooks, and tooling reduce chaos.

Learn from every incident

PIR for P1/P2 is required. Track action items. Prevent recurrence.